

Incident Response Report: Suspicious SSH Connection Attempts

Incident Overview

Date: September 28, 2025

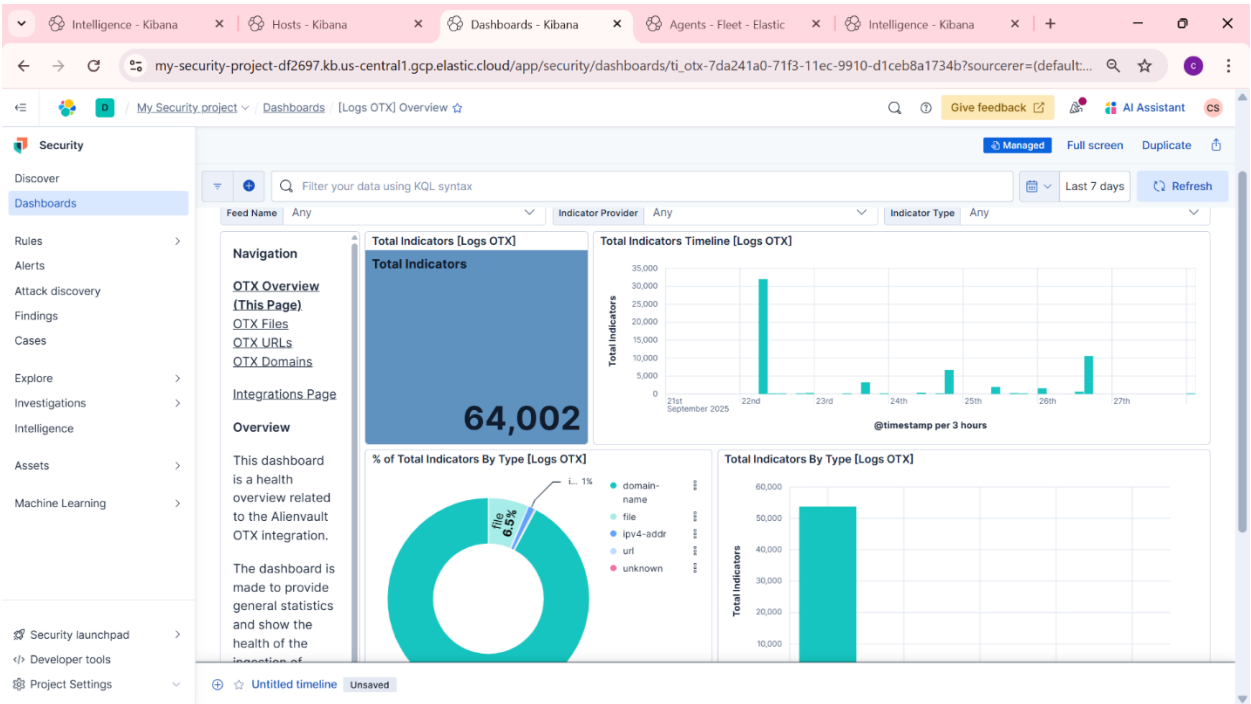
Time: 04:18:34 UTC

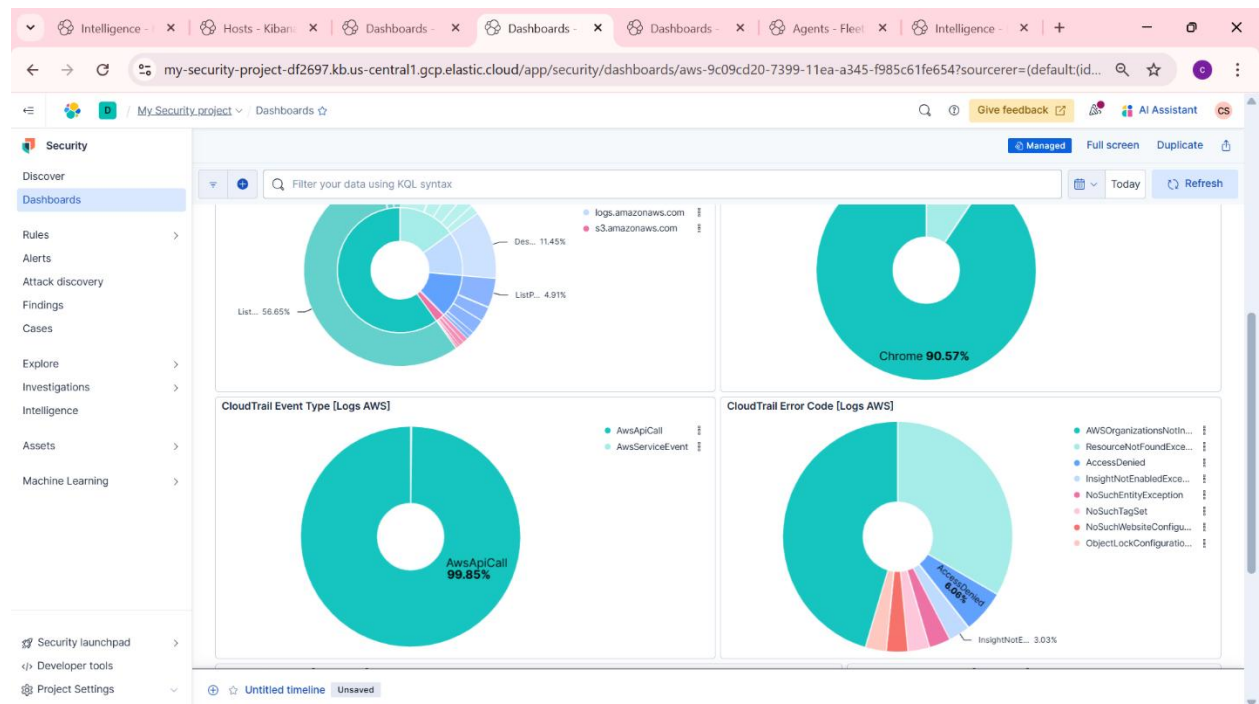
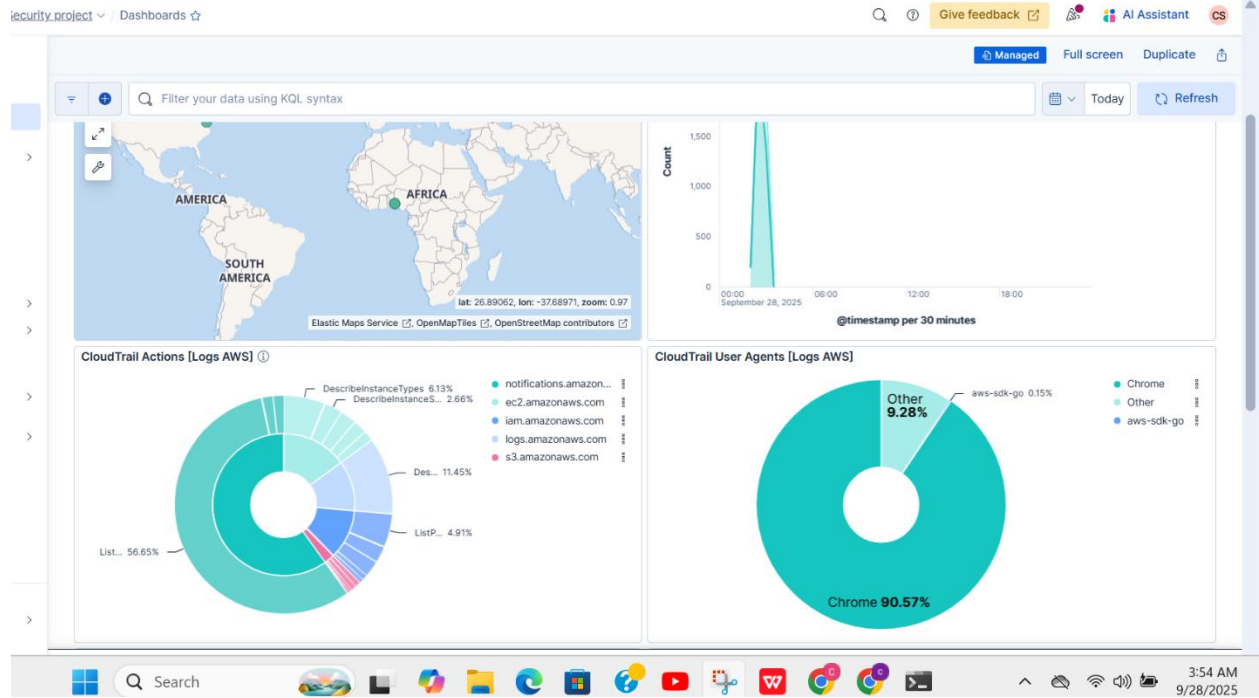
Severity: High

Status: Under Investigation

Executive Summary

On September 28, 2025, our Elastic Security solution detected multiple suspicious SSH connection attempts targeting one of our AWS EC2 instances in the us-east-1 region. The alert was triggered by our "SSH ATTEMPT DETECTION RULE" with a high-risk score of 88. This report documents our investigation findings and response actions taken to address this potential security incident.





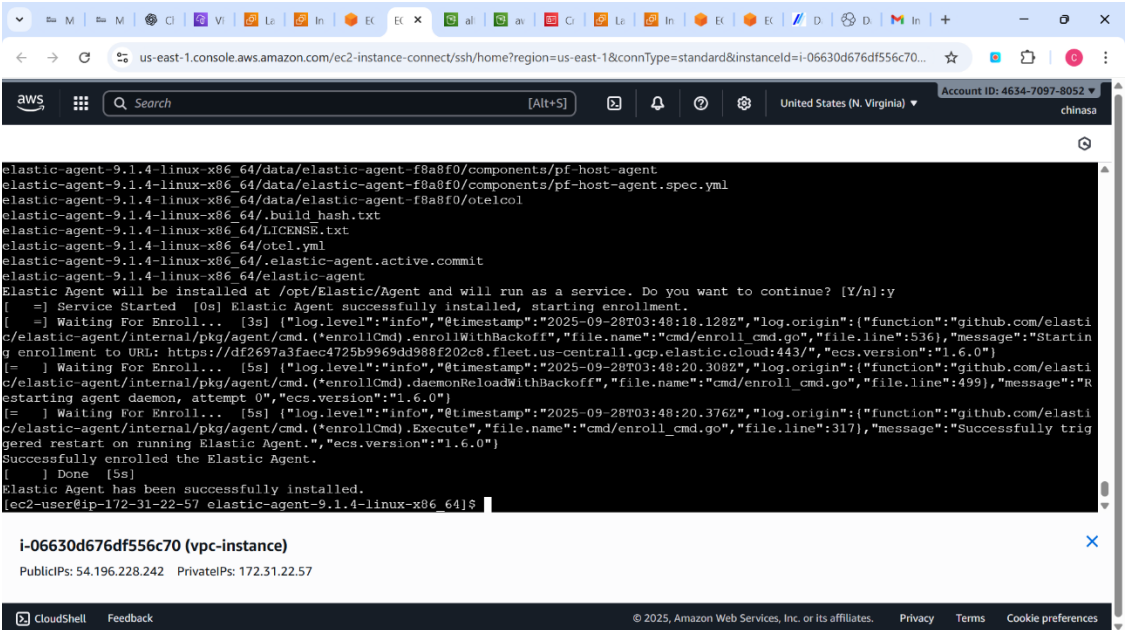
Incident Timeline

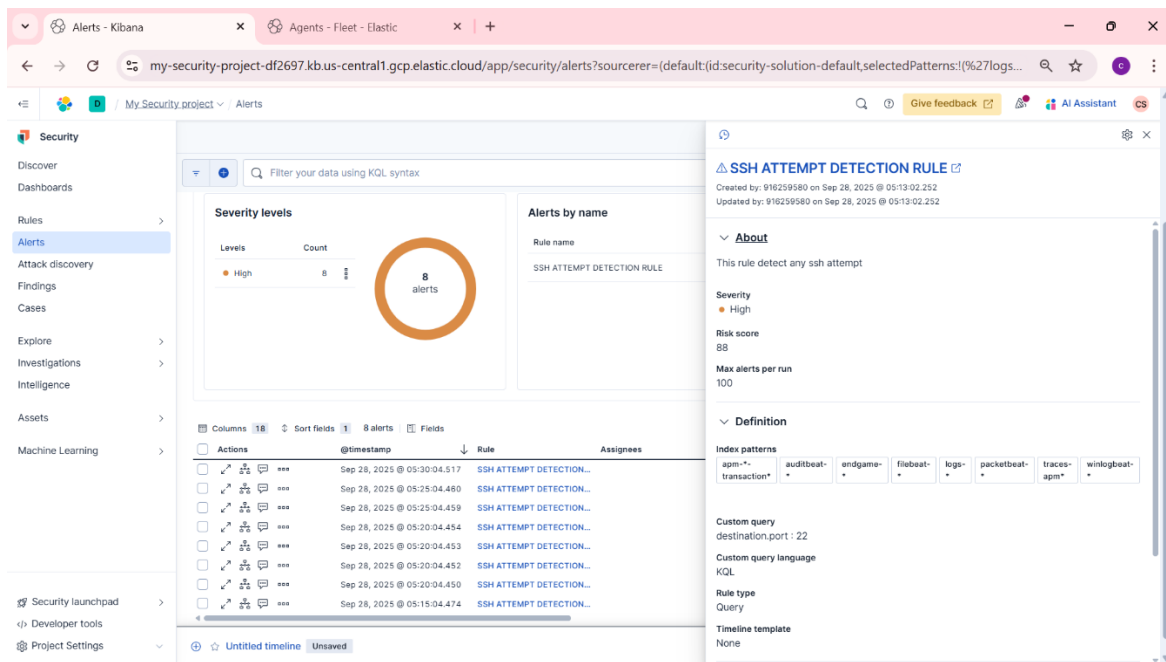
Time (UTC)	Event
2025-09-28 04:18:34	Initial SSH connection attempt detected from 18.206.107.29 to 172.31.22.57
2025-09-28 04:20:04	Alert generated in Elastic Security with risk score 88
2025-09-28 06:27:00	Security team notified and investigation initiated
2025-09-28 06:35:00	Decision made to isolate the host for further investigation
2025-09-28 06:40:00	Host successfully isolated using Elastic Defend

Attack Details

The suspicious activity involved SSH connection attempts from external IP address 18.206.107.29 to our EC2 instance (172.31.22.57). Our investigation revealed:

- Multiple connection attempts from the same source IP to different internal AWS instances
- Similar SSH attempts from related IP addresses (18.206.107.27 and 102.89.68.233)
- The source IPs appear to be from AWS infrastructure, potentially indicating a compromised EC2 instance being used as a launch point





Investigation Actions Taken

Initial Triage:

- Verified the EC2 instance should not be receiving external SSH connections
- Confirmed the source IP is not on our allow list
- Reviewed AWS CloudTrail logs for related suspicious activity

Host Isolation:

- Isolated the affected EC2 instance using Elastic Defend to prevent potential lateral movement
- Maintained monitoring capabilities while preventing outbound connections

Log Analysis:

- Reviewed SSH authentication logs on the target instance
- Analyzed VPC Flow Logs to identify the full scope of connection attempts
- Created a timeline in Elastic Security to visualize the attack pattern

Key Findings

- The SSH connection attempts appear to be part of a broader scanning campaign targeting multiple AWS instances
- Initial analysis indicates the attempts were unsuccessful due to our key-based authentication requirements

- No evidence of successful compromise was found on the affected instance
- The pattern matches known SSH brute force attempts seen in other AWS environments

Response Actions

Immediate Actions Taken:

- Isolated the host using Elastic Defend to prevent potential lateral movement
- Added the suspicious IPs to AWS Security Group deny rules
- Notified the cloud infrastructure team about the suspicious activity

Remediation Steps:

- Verified SSH configuration to ensure password authentication remains disabled
- Rotated SSH keys as a precautionary measure
- Updated Security Group rules to restrict SSH access to VPN IP ranges only
- Enhanced logging for SSH connection attempts

Monitoring Implemented:

- Created a custom watch list for the suspicious IP addresses
- Set up additional alerts for any SSH activity from external sources
- Scheduled a follow-up review in 24 hours to confirm no further suspicious activity

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- Lessons Learned & Recommendations

Security Improvements:

- Consider implementing a bastion host architecture for SSH access
- Move SSH to a non-standard port to reduce automated scanning exposure
- Implement just-in-time access for SSH connections to production instances

Process Improvements:

- Enhance alert correlation to identify related SSH scanning attempts more quickly
- Consider implementing automatic blocking for IPs making multiple failed SSH attempts

Next Steps

- Continue monitoring for 48 hours to ensure no further suspicious activity
- Release host from isolation after confirming no compromise (estimated: September 30)
- Schedule a security review of all EC2 instances to verify proper SSH configurations
- Update security awareness training to include this incident as a case study

Appendix

Related MITRE ATT&CK Techniques

T1021.004 - Remote Services: SSH

T1110 - Brute Force

T1190 - Exploit Public-Facing Application

Report prepared by: ALI CHINASA JULIET

Date: September 28 2025

Contact: chinasachinasa2014@gmail.com/cyberpracticals@gmail.com